Hand Gesture Based Augmented Reality System

Project Report

1. Introduction

The Hand Gesture Based Augmented Reality (AR) System is a computer vision project that enables users to interact with virtual objects using natural hand gestures. The system uses a standard webcam and real-time hand landmark detection to create an immersive AR experience without requiring external hardware such as gloves or sensors.

2. Objectives

- To detect and track hand gestures in real time.
- To enable interaction with virtual objects using gestures.
- To demonstrate the application of AI and computer vision in AR systems.

3. Technologies Used

- Python
- OpenCV
- MediaPipe
- NumPy

4. System Features

- Pinch gesture detection to spawn AR objects.
- 3D cube following finger movement.
- Hand-controlled AR menu.
- Live webcam-based interaction.

5. Future Scope

- Multi-hand support.
- Advanced gesture-based object manipulation.
- Integration with OpenGL for true 3D rendering.
- Voice and gesture hybrid interaction.
- Expansion into education, gaming, and smart interfaces.

6. Conclusion

This project demonstrates the effective use of hand gesture recognition and augmented reality concepts to create an interactive system. It highlights the potential of AI-driven interfaces in future human-computer interaction systems.